Phytoestrogens and their low dose combinations inhibit mRNA expression and activity of aromatase in human granulosa-luteal cells.

There is evidence that certain phytoestrogens inhibit aromatase, the enzyme that converts androgens to oestrogens. Kinetic studies in cell-free preparations show that they may inhibit aromatase by competitive binding to the enzyme, but there is a paucity of studies investigating longer-term effects of phytoestrogens on the expression of steroidogenic enzymes. This study tested the hypothesis that phytoestrogens could reduce aromatase activity by down-regulation of its expression. Experiments were carried out on primary cultures of human granulosa-luteal (GL) cells after they had been exposed to phytoestrogens for 48 h. Aromatase activity was measured by the ability of cells to convert testosterone to estradiol over a 4h period and aromatase mRNA expression (mRNA(arom)) was subsequently measured from the same cells using quantitative real-time PCR. The compounds investigated were the flavones, apigenin and quercetin, and the isoflavones, genistein, biochanin A and daidzein at doses of 10 microM and 100 nM. Combinations of these compounds at the lower dose were also investigated. All compounds tested dose-dependently reduced mean mRNA(arom) compared with controls. Apigenin was the most potent inhibitor with significant inhibition of mRNA(arom) seen at both 10 microM and 100 nM, whilst other flavonoids (except biochanin A) only induced significant inhibition (p<or=0.05) at the higher dose. Low dose (100 nM) mixtures of the compounds were ineffective except for a combination of the three isoflavones that induced a significant inhibition of mRNA(arom). The changes in aromatase activity paralleled the mRNA(arom) results and additional studies showed that the reduction in aromatase activity was significantly delayed in time compared with the reduction in mRNA(arom.) This is the first study to compare the action of various phytoestrogens, either singly or in low-dose combination, on the expression and activity of aromatase in human cells and it suggests that chronic dietary exposure and tissue accumulation of low-dose mixtures of phytoestrogens could have important consequences for aromatase activity and the production of oestrogens.